

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report:
Use a Port Scanner to Detect Open Ports

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OBJECTIVES

- Use Nmap, a port scanner and network mapping tool, to detect open ports.

BACKGROUND / SCENARIO

Network Mapper, or Nmap, is an open-source utility used for network discovery and security auditing. A common task is to scan local machines to determine potential vulnerabilities including open and unmanaged ports. All workstations require open ports and services to communicate and perform tasks like printing, sharing a file, or browsing the web. Administrators also use Nmap for monitoring hosts or managing service upgrade schedules. Nmap determines what hosts are available on a network, what services are running, what operating systems are running, and what packet filters or firewalls are running. In this lab, you will use Nmap inside your VM environment to detect open ports.

INTRODUCTION TO TCP/UDP PORTS

All communication that happens over the internet is exchanged using ports. Every IP host can use two types of ports: TCP and UDP. There can be up to 65,535 of each for any given IP address.

Services that connect to the internet use specific ports to receive information. The Internet Assigned Numbers Authority (IANA) divides ports into three sub-categories:

- Well-Known Ports (0 – 1023)
- Registered Ports (1024 – 49,151)
- Dynamic / Private Ports (49,152 – 65,535)

Common Port Reference

Port	Protocol	Description
20	FTP-DATA	File Transfer Protocol – Data
21	FTP	File Transfer Protocol – Control
22	SSH	Secure Shell
23	TELNET	Telnet
25	SMTP	Simple Mail Transfer Protocol
53	DNS	Domain Name System
67	DHCPv4	DHCP – Client to Server
68	DHCPv4	DHCP – Server to Client
69	TFTP	Trivial File Transfer Protocol
80	HTTP	Hypertext Transfer Protocol

SECURITY OF LOGICAL PORTS

Every logical port is subject to a threat and poses a vulnerability to a system. Over 75% of all cyberattacks involve just a few common ports. Attackers scan systems to identify open ports on a target system. Below is a placeholder for the list of commonly targeted ports:

Here are some **commonly targeted ports** by hackers:

- **Port 22 (SSH):** Used for secure remote access, often targeted for brute-force attacks. [↔ 2](#)
- **Port 80 (HTTP):** The standard port for unencrypted web traffic, frequently exploited for credential theft. [↔ 2](#)
- **Port 443 (HTTPS):** Used for secure web traffic, also a common target for exploits. [↔ 2](#)
- **Port 3389 (RDP):** Used for Remote Desktop Protocol, a primary target for ransomware attacks. [↔ 2](#)
- **Port 25 (SMTP):** Used for email transmission, often targeted for data breaches. [↔ 2](#)

Understanding these ports can help enhance your network security and protect against potential vulnerabilities.

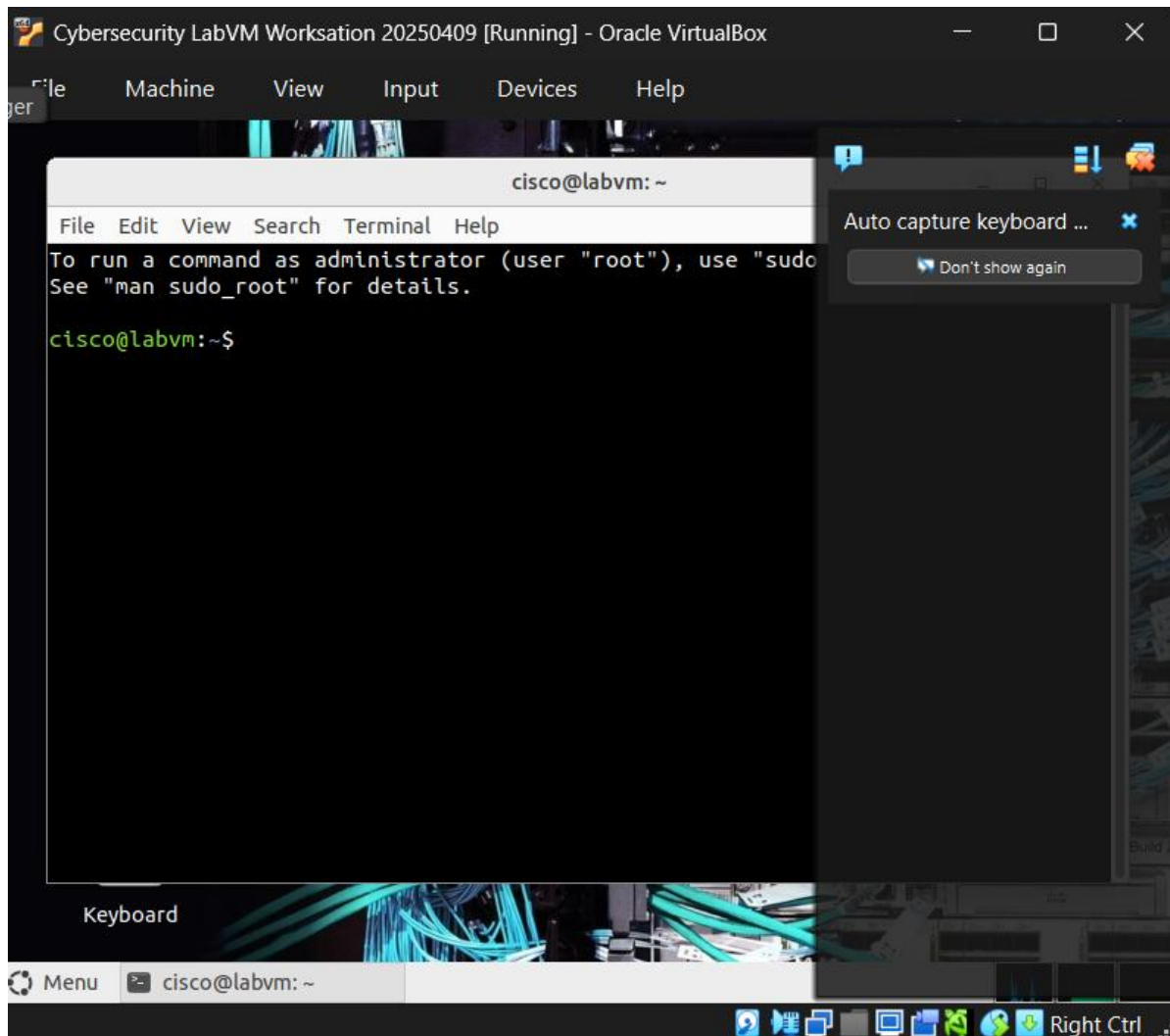
REQUIRED RESOURCES

- PC with the CSE-LABVM installed in VirtualBox.

INSTRUCTIONS

Step 1: Open a Terminal Window in the CSE-LABVM

- Launch the CSE-LABVM.
- Double-click the Terminal icon to open a terminal.



Step 2: Run Nmap

At the command prompt, enter the following command to run a basic scan:

```
cisco@labvm:~$ nmap localhost
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-02 11:27 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00032s latency).
Not shown: 996 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
631/tcp   open  ipp

Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds
cisco@labvm:~$
```

The results show a scan of the first 1024 TCP ports.

Step 3: Use Administrative Privileges with Nmap

a. Scan UDP ports using sudo:

```
cisco@labvm:~$ sudo nmap -sU localhost
[sudo] password for cisco:
Sorry, try again.
[sudo] password for cisco:
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-02 11:38 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000017s latency).
Not shown: 997 closed ports
PORT      STATE      SERVICE
123/udp    open       ntp
631/udp    open|filtered ipp
5353/udp   open|filtered zeroconf

Nmap done: 1 IP address (1 host up) scanned in 1.33 seconds
```

b. Run a version detection scan:

```
cisco@labvm:~$ nmap -sV localhost
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-02 11:51 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00015s latency).
Not shown: 996 closed ports
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.0.8 or later
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.1 (Ubuntu Linux; protocol 2.0)
23/tcp    open  telnet   Linux telnetd
631/tcp   open  ipp?
Service Info: Host: Welcome; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 189.43 seconds
```

The -sV switch performs version detection, which can be used to research vulnerabilities.

Step 4: Capture SSH Keys

a. Run an aggressive script scan:

```
cisco@labvm:~$ nmap -A localhost
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-02 11:56 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00011s latency).
Not shown: 996 closed ports
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.0.8 or later
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 127.0.0.1
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   At session startup, client count was 2
|   vsFTPD 3.0.5 - secure, fast, stable
|_End of status
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.1 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 be:7d:82:5e:38:db:f3:54:a3:cd:75:b2:ff:b5:ce:29 (ECDSA)
|_  256 12:e2:5b:5c:ef:f1:35:eb:9e:8f:a2:2f:39:de:94:18 (ED25519)
23/tcp    open  telnet   Linux telnetd
631/tcp   open  ipp?
Service Info: Host: Welcome; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 238.46 seconds
```

The -A flag runs NSE scripts to test specific vulnerabilities and capture details such as SSH host keys.

b. View the Nmap manual pages:

```
NMAP(1)                                Nmap Reference Guide                                NMAP(1)

NAME
    nmap - Network exploration tool and security / port scanner

SYNOPSIS
    nmap [Scan Type...] [Options] {target specification}

DESCRIPTION
    Nmap ("Network Mapper") is an open source tool for network exploration
    and security auditing. It was designed to rapidly scan large networks,
    although it works fine against single hosts. Nmap uses raw IP packets
    in novel ways to determine what hosts are available on the network,
    what services (application name and version) those hosts are offering,
    what operating systems (and OS versions) they are running, what type of
    packet filters/firewalls are in use, and dozens of other
    characteristics. While Nmap is commonly used for security audits, many
    systems and network administrators find it useful for routine tasks
    such as network inventory, managing service upgrade schedules, and
    monitoring host or service uptime.

    The output from Nmap is a list of scanned targets, with supplemental
    information on each depending on the options used. Key among that

Manual page nmap(1) line 1 (press h for help or q to quit)
```

Enter q or quit to exit the man pages at any time.

SUMMARY – HIGHLY VULNERABLE PORTS

Many ports must be open for a host to function in a normal computing and communication environment. However, these ports should be monitored regularly to ensure they are not compromised.

TCP Port 21 – FTP

- Unauthorized transfer, deletion, and modification of files
- Unauthorized transfer of malicious code or payloads
- Anonymous authentication to host file systems
- Injection of malicious scripts (e.g., XSS attacks)
- Impact the availability of other host services

TCP Port 23 – Telnet

- Gain unauthorized remote access to a host
- Plant backdoors and other types of malicious code
- View sensitive data and credentials (data is transmitted in plaintext)
- Perform man-in-the-middle attacks
- Impact the availability of other host services

TCP Port 53 – DNS

- Most common exit route for attackers after a breach
- Rarely monitored — attackers use it to exfiltrate data and cover tracks
- Used to clear files, logs, and other evidence

TCP Port 80 – HTTP

- Unauthorized transfer, deletion, and modification of data
- Unauthorized transfer of malicious code or payloads
- Injection of malicious scripts (e.g., XSS attacks)
- Impact the availability of other host services